

Estimation and Evaluation of Reference Evapotranspiration Using an AI-Agent-Based Automated Meteorological Data System

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Abstract

This study presents an automated meteorological data management system with an AI agent to calculate ET_o by applying the standard FAO-56 Penman Monteith (PM) equation. The proposed framework automatically performs meteorological data acquisition, quality control (QC), management, ET_o calculation, aggregation, visualization, and statistical evaluation. This study used daily meteorological data from 21 Automated Synoptic Observing System (ASOS) stations in the Nakdong River Basin, Republic of Korea, covering 2011 to 2023, and integration of API-retrieved meteorological data began from 2024 onward. The automated ET_o results were compared with conventionally calculated PM-based potential evapotranspiration (PET) using RMSE, MAE, PBIAS, and R2. Findings are consistent across all stations, with monthly RMSE mostly less than 0.7 mm (maximum 1.14 mm), MAE less than 0.7 mm, PBIAS in the range -0.465-0.833%, and R2 in the range 0.999-1.000. The results indicate that the automated workflow maintained the physical consistency of ET_o estimation and improved reproducibility and applicability. The proposed system may support operational agricultural water management in the Nakdong River Basin.

Key words – AI-Agent, ASOS meteorological data, Penman-Monteith, Reference evapotranspiration.

Introduction

Precise estimation of reference evapotranspiration (ET_o) is a key requirement to ensure agricultural production and the efficient use of water resources, especially in areas where climate variability is increasing (Waqas & Kim, 2026a). Meteorological parameters such as temperature, precipitation, and extreme weather events significantly affect ET_o and water demand in the agricultural sector (Koç & Can, 2023; Sentelhas et al., 2010; Waqas & Kim, 2026a). The effects of these changes are particularly pronounced in basins with climatic diversity, which is why the Nakdong River Basin in South Korea can be viewed as a highly complex topography subject to monsoon-based climate patterns that introduce substantial spatial and temporal variability in meteorological conditions (Bidabadi et al., 2024; Waqas & Kim, 2026b). Under such circumstances,

long-term observation and precise estimation of ET_o are necessary to develop adaptive irrigation plans.

Many empirical and physically based methods have been developed for ET_o estimation. The most commonly adopted FAO-56 Penman Monteith (PM) equation, which is internationally recognized as the standard method due to its physical basis and good performance under different climatic conditions (Allen et al., 1998). Many validation studies have shown that the relationship between the FAO-56 PM method and lysimeter measurements is similar to the other empirical methods, including the Hargreaves-Samani, Blaney-Cridle, and Priestley-Taylor methods (Alam et al., 2024; Tabari et al., 2013; Trajkovic & Kolakovic, 2009; Wanniarachchi & Sarukkalgige, 2022). Accordingly, FAO-56 PM has gained status as a reference framework for research and operational applications.

In the Republic of Korea, to obtain high-frequency, long-term

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measurements of meteorological parameters, the Korea Meteorological Administration (KMA, 2026) employs the Automated Synoptic Observing System (ASOS). Archived ASOS data can be analyzed using historical trend analysis, and current observations can be continuously added to analysis systems via application programming interfaces (APIs). However, traditional manual or semi-automatic processing procedures cannot effectively handle such a large volume of data or provide consistency, scale, and repeatability (Samani, 2000; Wanniarachchi & Sarukkalgige, 2022).

Intelligent automation and artificial intelligence (AI) are among the technologies that have emerged as remedies to these shortcomings over the last few years (Fang et al., 2024; Goyal et al., 2023). The use of AI-based algorithms to optimize the accuracy of estimates or physically based models in the case of data scarcity has been actively researched in the context of ET_o (El-Kenawy et al., 2022; Farag, 2025; Waqas & Kim, 2026a). Though this has enhanced predictive performance, little has been done to automate the end-to-end ET_o estimation workflow.

Current literature is split-process-based, where meteorological data are acquired and pre-processed, ET_o is calculated, and the results are visualized in separate tools and manually processed. These methods limit scalability and real-time utility

and, in most cases, lead to discrepancies across research in similar geographical regions (Goyal et al., 2023; Sentelhas et al., 2010). Moreover, data QC procedures are often not documented and are specific to a particular researcher, thereby reducing transparency and reproducibility (Padhiary et al., 2025; Ye et al., 2024). These restrictions delay the use of ET_o estimation in operational decision-support systems for agricultural and water resources management.

This study proposes a rule-based automation framework using an AI agent for acquiring and pre-processing meteorological data, estimating ET_o using the FAO-56 method, and visualizing and evaluating the results within a single end-to-end workflow using ASOS data. The proposed system improves the scalability, repeatability, and operability of water management and climate change adaptation in the agricultural domain of the Nakdong River Basin, compared to conventional split-process approaches.

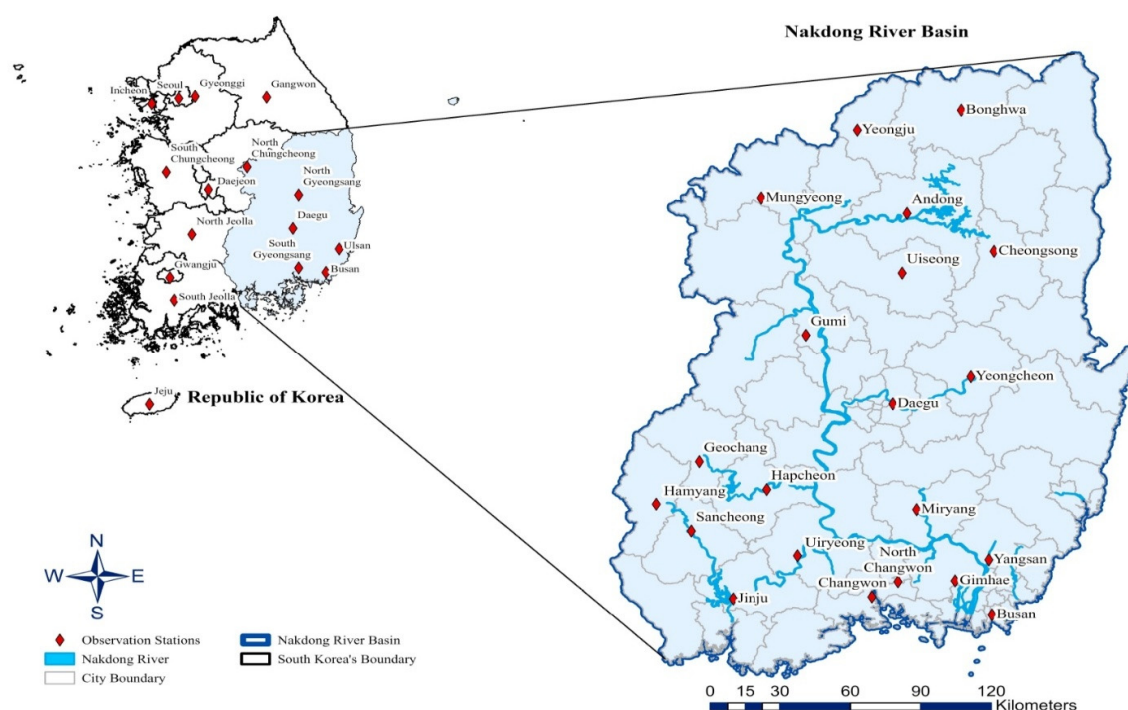


Fig. 1. Locations of the 21 ASOS meteorological stations within the Nakdong River Basin.

Materials and Methods

1. Study area

This study was conducted in the Nakdong River Basin (Fig. 1), one of the largest and most hydrologically significant river basins in the Republic of Korea. The basin is in the southern section of the Korean Peninsula and spans about 23,000 km² (Waqas & Kim, 2026a). The Nakdong River Basin climate is temperate monsoon, influenced by the East Asian summer monsoon. The rainfall averages 1,000 to 1,600 mm/year, with most falling in the summer months (June to September). The effect of this pattern of seasonal rainfall distribution is severe intra-annual variation in the hydrological cycle, including ET (Chung et al., 2004).

2. Data acquisition and quality checking

The meteorological data used in this study were collected from 21 ASOS stations managed by the KMA(2026).

The ASOS system delivers real-time, standardized measurements of the most critical meteorological parameters needed

to estimate ET_o using the FAO-56 PM method: maximum and minimum air temperature, wind velocity, relative humidity, and proxies of solar radiation. In this study, two supplementary data sources were combined. To begin, a historical dataset (2011-2023) was imported from standardized Excel files. Second, the KMA ASOS Daily API was used to retrieve meteorological observations for 2024 and later automatically. Retrieved variables were transformed into a unified, structure-compatible format with the historical dataset.

An extra QC step was added in the pre-processing stage to enhance the reliability of FAO-56 PM calculations. Structural validation of station IDs, dates, and variable completeness, and physical consistency screening of meteorological variables were performed during the QC process. The unrealistic extremes and inconsistencies in $T_{\min}$ - $T_{\max}$ were checked. Relative humidity values outside the physically acceptable range (0-100%) were not included. Data on wind speed were checked for negative or abnormal values and converted to the standard 2 m wind speed (u_2) required by the FAO-56 PM equation, assuming a wind measurement height

Table 1. Metadata of the 21 ASOS stations, including latitude, longitude, and elevation

Station	Station number	Latitude	Elevation(m)	Longitude
Geochang	284	35.66739	228.45	127.9099
Gimhae	253	35.22981	54.59	128.8908
Miryang	288	35.49147	8.31	128.7441
North Changwon	255	35.22655	50.95	128.6726
Sancheong	289	35.413	138.22	127.8791
Yangsan	257	35.30737	6.29	129.0201
Uiryeong	263	35.32258	14.1	128.2881
Jinju	192	35.16378	29.35	128.04
Changwon	155	35.17019	34.97	128.5728
Hamyang	264	35.51138	152.07	127.7454
Hapcheon	285	35.56505	26.72	128.1699
Gumi	279	36.13055	49.17	128.3206
Mungyeong	273	36.62727	173.01	128.1488
Bonghwa	271	36.94361	324.67	128.9145
Andong	136	36.57293	141.26	128.7073
Yeongju	272	36.87183	211.32	128.5169
Yeongcheon	281	35.97742	96.12	128.9514
Uiseong	278	36.3561	81.511	128.6886
Cheongsong	276	36.4351	208.65	129.0401
Busan	159	35.10468	69.56	129.032
Daegu	143	35.87797	54.27	128.653

of 10 m: $u_2 = u_h \times 4.87 / \ln(67.8h - 5.42)$, where u_h is the wind speed measured at height h . Solar radiation proxy variables and sunshine-duration-related parameters were tested to identify physically unrealistic radiation estimates and outliers. Records with missing data were left as NaN to maintain temporal continuity, while records required to calculate FAO-56 PM were automatically excluded from the ET_0 estimation.

The historical and API data were then pre-processed and merged to form a continuous time series for each station. The built-in QC and pre-processing system enabled automatic identification, filtering, and standardization of meteorological observations prior to ET_0 calculation. It is a unified, automated data processing framework that ensures the continuity, consistency, and reliability of ET_0 estimation across all observation stations.

3. Methodology

3.1. AI-agent-based ET_0 estimation framework

This study uses an AI-agent-based computational system to automatically calculate, compare, and analyze ET_0 after quality-checking the input weather data. The framework combines

rule-based data ingestion, QC, physical modeling, aggregation, visualization, and performance-evaluation agents (Fig. 2). The proposed framework is rule-based automation and deterministic physical modeling, without machine learning. The AI agent acts as an orchestration and decision-making layer that understands the user's request, checks for required inputs, fetches the data, applies pre-processing and QC, calculates ET_0 , and generates automatically summarized outputs. The AI agent uses natural-language input from the user to identify target stations and times, automatically fetches meteorological data from multiple sources, computes daily ET_0 , collates results monthly, and compares the automated results with conventionally calculated FAO-56 PM-based PET values to assess computational consistency. The system also uses a set of rules to recognize the intent and extract parameters from natural language queries, such as station ID, region, date range, and meteorological parameters. The workflow in Fig. 2 follows a sequential decision-based architecture comprising input validation, parameter verification, API request generation, data retrieval, pre-processing, ET_0 computation, and output generation modules. There are two data collection methods. ASOS meteorological data that have been pre-processed span 2011-2023. The KMA-ASOS API enables the agent to collect

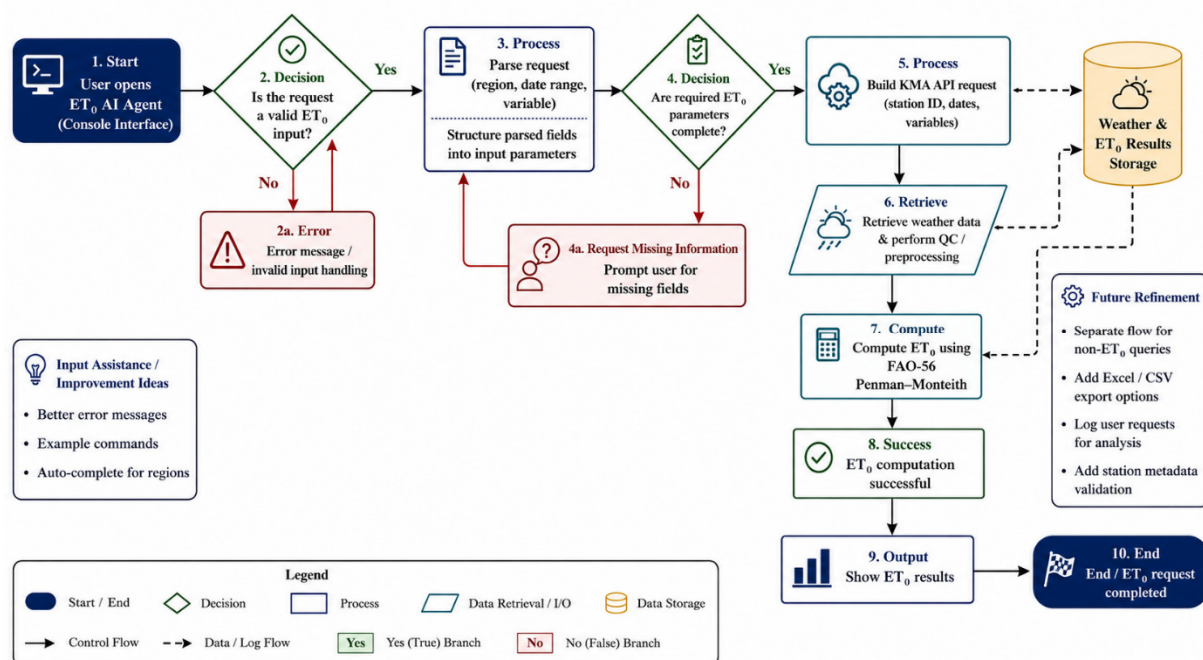


Fig. 2. Overall workflow of the AI-agent-based automated ET_0 estimation system.

the data from 2024 onward. The data from these two sources are integrated into a single dataset, and ET_0 is computed after screening for simple structural consistency in date order, variable structure, and units. The pre-processing and QC procedures involve checking for chronological consistency, missing or duplicate records, variable structure consistency, and unit standardization prior to model execution.

The ET_0 calculation module is not a data-driven learning model but a physically based FAO-56 PM equation. The agent automatically generates the meteorological data required for the calculation, namely the air temperature, relative humidity, wind speed, and solar radiation variables, and then performs standard FAO-56 calculations to determine the daily reference evapotranspiration. Then, monthly aggregation and statistical validation are carried out via automated evaluation routines. The novelty of the framework lies in the autonomous integration of natural-language interaction, rule-based workflow management, automatic acquisition of meteorological data, quality-controlled pre-processing, and standardized ET_0 calculation within a single AI-agent environment.

3.2. FAO-56 Reference PM method

ET_0 was calculated using the FAO-56 Penman-Monteith equation, which is recommended as the standard method for estimating ET_0 under a wide range of climatic conditions (Allen et al., 1998). Due to the lack of direct lysimeter observations and long-term official reference ET_0 datasets for the study region, the FAO-56 PM method was used as the computational framework for this study. Hence, the comparison was conducted to assess the computational consistency and reproducibility of the automated meteorological data acquisition and processing workflow by comparing its results with those from a conventionally implemented FAO-56 PM calculation using the same meteorological inputs. This comparison enabled the evaluation of whether the automated workflow could reproduce the ET_0 values obtained through the traditional manual FAO-56 PM calculation procedure.

The daily ET_0 was computed as follows:

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

Where ET_0 is the reference evapotranspiration (mm day^{-1}), Δ

is the slope of the saturation vapor pressure curve ($\text{kPa } ^\circ\text{C}^{-1}$), R_n is the net radiation at the crop surface ($\text{MJ m}^{-2} \text{ day}^{-1}$), G is the soil heat flux density ($\text{MJ m}^{-2} \text{ day}^{-1}$), γ is the psychrometric constant ($\text{kPa } ^\circ\text{C}^{-1}$), T is the mean daily air temperature at 2 m height ($^\circ\text{C}$), u_2 is the wind speed at 2 m height (m s^{-1}), e_s is the saturation vapor pressure (kPa), e_a is the actual vapor pressure (kPa), and $e_s - e_a$ is the saturation vapor pressure deficit (kPa).

3.3. Statistical evaluation criteria

To assess the computational consistency of the AI-agent-based ET_0 calculation workflow, the calculated ET_0 values were systematically compared with PET values obtained using the conventional FAO-56 PM implementation. The comparison was not intended to validate ET_0 independently but to assess the computational consistency, data-processing reliability, and reproducibility of the automated workflow, because both datasets were based on the same theoretical FAO-56 PM framework. The statistical analysis thus compared the extent of agreement between the automated AI-agent workflow and the standard FAO-56 PM calculation procedure. Further strengthening of external validation across varying climatic conditions could be achieved by future studies that include independent field observations, such as lysimeter measurements or officially validated ET_0 datasets. Four complementary statistical measures were used: root mean square error (RMSE), mean absolute error (MAE), percent bias (PBIAS), and coefficient of determination (R^2).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (ET_{0,i} - PET_i)^2} \quad (2)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |ET_{0,i} - PET_i| \quad (3)$$

$$\text{PBIAS}(\%) = 100 \times \frac{\sum_{i=1}^n (ET_{0,i} - PET_i)}{\sum_{i=1}^n PET_i} \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (PET_i - ET_{0,i})^2}{\sum_{i=1}^n (PET_i - \overline{PET})^2} \quad (5)$$

Results

Daily ET_o time series from the automated meteorological data system, powered by an AI agent, exhibit high seasonal coherence, interannual consistency, and strong agreement between historical data and API-retrieved ASOS data across typical climatic conditions in the Nakdong River Basin.

A strong annual cycle is observed in daily ET_o at Jinju station, with winter lows typically ranging from about 0.6 to 1.0 mm/day, lower than the summer highs, which are often over 5.0 mm/day and, in exceptional cases, 6.0 mm/day (Fig. 3a).

Busan station shows a different but consistent ET_o regime due to its coastal location (Fig. 3b). Daily ET_o is higher than at Jinju in both summer and winter. Winter ET_o usually does not fall below 1.0 mm/day; summer values are often over 6.0

mm/day, and sometimes up to 7.0 mm/day. This increase in evaporation can be explained by higher average air temperatures, stronger coastal winds, and humidity gradients that characterize the southern coastal belt. Although the variation within a shorter band is more substantial, especially in summer seasons when synoptic weather systems heighten wind and radiation conditions, the moving average shows a consistent, reproducible seasonal pattern across all four seasons.

The findings in Fig. 3 indicate that the proposed workflow generated continuous and seasonally consistent ET_o estimates at the selected stations, suggesting its applicability to ET_o monitoring in the Nakdong River Basin.

PBIAS in the basin range from 0.833 to -0.465, indicating low systematic bias, slightly overestimated or underestimated by location (Table 2). Notably, none of the stations have an

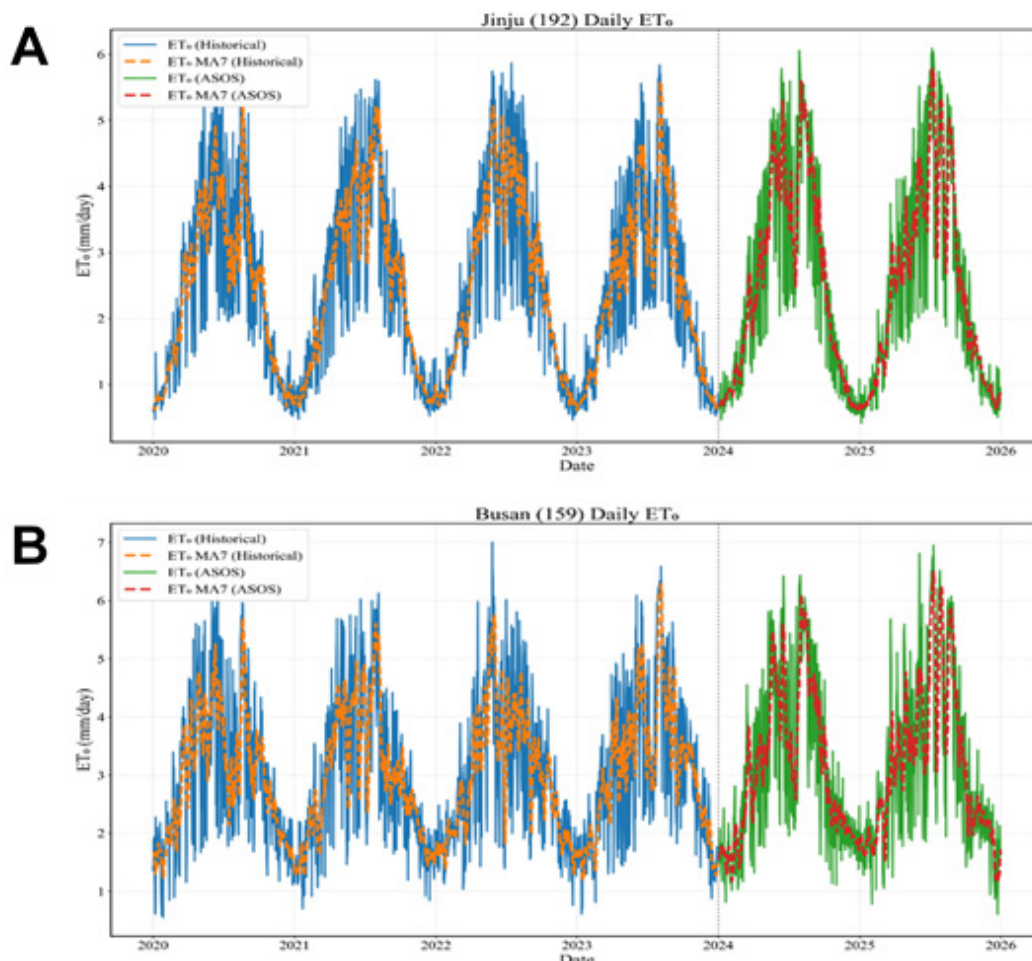


Fig. 3. Daily estimated ET_o at the Jinju and Busan stations for the period 2020–2025 (a) Jinju station, and (b) Busan station. A: Daily ET_o at the Jinju station (192) for the period 2020–2025, B: Daily ET_o at the Busan station (159) for the period 2020–2025.

Table 2. Monthly verification statistics of ET_0 and PET at 21 sites in the Nakdong River basin (2011–2023)

Station	PBIAS (%)	RMSE (mm)	MAE (mm)	R^2
Daegu	0.002	0.003	0.002	1.000
Yeongcheon	0.018	0.068	0.059	1.000
Gumi	-0.284	0.241	0.218	1.000
Hapcheon	0.259	0.248	0.209	1.000
Miryang	0.272	0.295	0.249	1.000
Cheongsong	-0.029	0.323	0.288	1.000
Uiseong	-0.416	0.366	0.310	1.000
Uiryeong	0.479	0.449	0.376	1.000
Yangsan	0.426	0.471	0.382	1.000
Mungyeong	-0.391	0.471	0.410	1.000
Andong	-0.423	0.553	0.437	1.000
Sancheong	0.709	0.575	0.555	1.000
Yeongju	-0.465	0.577	0.502	1.000
Bonghwa	-0.119	0.610	0.550	1.000
Northchangwon	0.690	0.618	0.541	1.000
Geochang	0.833	0.625	0.613	1.000
Changwon	0.702	0.676	0.577	1.000
Jinju	0.761	0.679	0.577	1.000
Hamyang	0.788	0.681	0.574	1.000
Busan	0.720	0.754	0.669	0.999
Gimhae	0.501	1.137	0.661	0.999

absolute PBIAS exceeding 1%, highlighting the numerical consistency of the automated pipeline. There is also a consistently low RMSE, ranging from 0.003 mm at Daegu to 1.137 mm at Gimhae, with the majority below 0.7 mm. MAE also shows consistently low values, with most stations reporting less than 0.7 mm.

On a smaller scale, slightly higher errors have been recorded at certain coastal and low-lying stations, e.g., Busan and Gimhae, where the RMSE is close to or slightly greater than 1 mm. It is a manifestation of radiation- and humidity-sensitive calculations of small anomalies in sunshine duration. However, at these stations, the relative error is small compared to monthly ET sums, and the R^2 is 0.999, which substantiates strong temporal consistency.

Table 2 shows that the automated workflow reproduced the PM-based PET values with high computational consistency at the monthly level. The low bias, low absolute error, and near-unity R^2 across the stations indicate the reproducibility and numerical stability of the proposed workflow.

Discussion

The quantitative assessment indicates that monthly ET_0 from the automated system is similar to PM-based PET, with most RMSE values below 0.7 mm and R^2 values of 0.999–1.000. These results suggest that the automated workflow maintained consistency with the conventional FAO-56 PM calculation and improved processing reproducibility. Monthly ET_0 is often more than 120–150 mm in inland and coastal South Korea. In such a case, the absolute deviations in this study are relative errors much lower than 1%, which are relatively small for monthly-scale applications, drought evaluation, and basin-scale water-balance modeling. Equivalent performance thresholds were found to be comparable in Korean lysimeter-calibrated or PM-based reference dataset studies of ET (Lee et al., 2023; Park et al., 2021).

The spatial pattern of errors at the 21 ASOS stations reveals distinct climatological characteristics. The RMSE at coastal stations such as Busan and Gimhae is slightly higher (0.75–1.14 mm) than at inland and mountainous stations. This kind of behavior

is consistent with earlier Korean literature, which revealed greater uncertainty in estimating ET_0 under maritime conditions, with stronger wind regimes and greater atmospheric humidity variability (Jeon et al., 2022; Kim & Kang, 2024).

Stations such as Daegu, Andong, and Gumi showed very high agreement between automated ET_0 and PET, with an RMSE of 0.003 -0.55 mm. These findings are consistent with studies that found FAO-56 PM-based ET_0 estimates in inland river basins in Korea are highly numerically stable when meteorological inputs are complete and well-controlled (An & Ha, 2026; Waqas & Kim, 2026a).

The proposed framework is not aimed at introducing a new empirical or machine-learning-based ET_0 estimation equation, but rather at developing an AI-agent-based automated system for implementing the standardized FAO-56 PM method. Unlike previous studies that mainly focused on improving ET_0 prediction accuracy using machine learning or deep learning techniques (Lee, 2020; Waqas & Kim, 2026a), the present study highlights that operational automation, intelligent workflow management, and integration of large meteorological datasets are highlighted, while maintaining the physical consistency and interpretability of the FAO-56 PM approach.

The main contribution of this study lies in integrating AI-agent-based decision logic with automated retrieval, QC, parameter validation, multi-station processing, and ET_0 computation of meteorological data within a single operational framework. Existing data automation studies generally focus on simple data collection or pre-processing pipelines. In contrast, the proposed system performs end-to-end automated task execution, including request validation, missing-parameter detection, automated API communication, weather-data pre-processing, FAO-56 PM computation, and real-time result generation.

A drawback of this study is that validation was performed against conventionally applied FAO-56-based PET calculations using the same meteorological inputs, rather than independent field measurements, such as lysimeter measurements. Further, the proposed framework was tested only in the Nakdong River Basin, so it needs to be tested under other climatic and hydrological conditions to ensure its wider generalizability and applicability.

Taken together, the AI-agent-based framework provided a reliable and reproducible operational workflow for estimating

ET_0 using quality-controlled ASOS meteorological data and the FAO-56 PM equation. The close agreement with conventionally computed PM-based PET across 21 stations indicates that workflow automation did not compromise the physical consistency or numerical stability of ET_0 estimation. Therefore, the proposed system may serve as a practical tool for supporting basin-scale irrigation scheduling, drought assessment, and agricultural water management in South Korea. Future studies should further validate the framework using independent observational datasets, such as lysimeter measurements or officially validated ET_0 data, and test its applicability under broader climatic and hydrological conditions.

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